

Section 4.4 l'Hospital's Rule

Do you remember how to calculate limits?

$$\underline{\text{Ex}} \quad \textcircled{1} \quad \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} \rightarrow \frac{2^3 - 8}{2^2 - 4} = \frac{0}{0}$$

Now factor and then reduce.

$$\lim_{x \rightarrow 2} \frac{(x-2)(x^2+2x+4)}{(x-2)(x+2)}$$

$$= \lim_{x \rightarrow 2} \frac{x^2+2x+4}{x+2}$$

$$= \frac{2^2 + 2 \cdot 2 + 4}{2+2}$$

$$= \frac{12}{4} = \textcircled{3}$$

What about using derivatives?

l'Hospital's Rule

If $\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$ is of the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

$$\text{then } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Ex ② $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$ is of the form $\frac{0}{0}$

l'Hospital's Rule $= \lim_{x \rightarrow 2} \frac{3x^2}{2x}$

$$= \frac{3(2)^2}{2 \cdot 2} = \textcircled{3}$$

NOTE Please write "l'Hospital's Rule"
or "l'H" each time you use it.

Ex ③ $\lim_{x \rightarrow \infty} \frac{4x^3 - 7x^2 + 9}{2x^3 + 3x^2 + x}$ is of the form $\frac{\infty}{\infty}$

$\text{L'H} = \lim_{x \rightarrow \infty} \frac{12x^2 - 14x}{6x^2 + 6x + 1}$ is of the form $\frac{\infty}{\infty}$

$\text{L'H} = \lim_{x \rightarrow \infty} \frac{24x - 14}{12x + 6}$ is of the form $\frac{\infty}{\infty}$

$\text{L'H} = \lim_{x \rightarrow \infty} \frac{24}{12} = \textcircled{2}$

EX ④ $\lim_{x \rightarrow 0} \frac{\sin x}{x}$ is of the form $\frac{0}{0}$

$\text{L'H} = \lim_{x \rightarrow 0} \frac{\cos x}{1}$

$= \cos(0) = \textcircled{1}$

Ex ⑤ $\lim_{x \rightarrow \infty} \frac{\sqrt{x} + 8}{x^2 + 1}$ is of the form $\frac{\infty}{\infty}$

$\text{L'H} = \lim_{x \rightarrow \infty} \frac{\frac{1}{\sqrt{x}}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x\sqrt{x}} = \frac{1}{\infty} \rightarrow \boxed{0}$

Ex ⑥ $\lim_{x \rightarrow 0} \frac{e^{-x} - 1}{\tan x}$ is of the form $\frac{0}{0}$

$$l'H = \lim_{x \rightarrow 0} \frac{e^{-x}(-1)}{\sec^2 x}$$

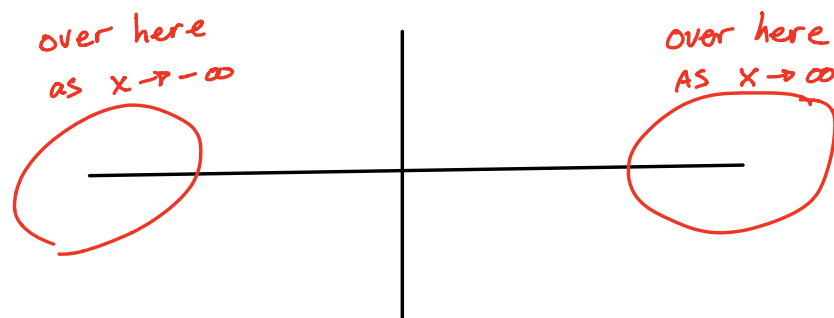
$$= \frac{e^{-0}(-1)}{\sec^2(0)} = \boxed{-1}$$

What does this have to do with the graphs of functions?

Ex ⑦ Describe the end behavior of

$$f(x) = \frac{8x^2 - 7x}{2x^2 + 4}$$

What happens on the far right side
And far left side of the graph?

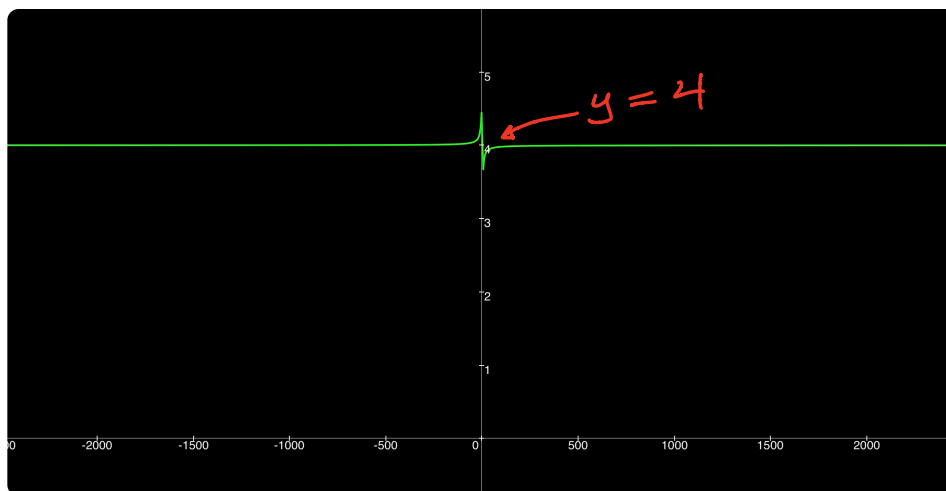


$$\lim_{x \rightarrow \infty} \frac{8x^2 - 7x}{2x^2 + 4}$$

$$1'H = \lim_{x \rightarrow \infty} \frac{16x - 7}{4x}$$

$$1'H = \lim_{x \rightarrow \infty} \frac{16}{4} = 4 \quad (\text{we get the same answer for } x \rightarrow -\infty)$$

So the ends of the graph level off at $y = 4$.



Ex ⑧ $\lim_{x \rightarrow 0^+} x \cdot \ln x$ is of the form $0 \cdot (-\infty)$,
we need $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Turn this into a rational function.

$$\lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \text{ is of the form } \frac{\infty}{\infty}$$

$$\text{L'H} = \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow 0^+} -\frac{x^2}{x}$$

$$= \lim_{x \rightarrow 0^+} -x = \boxed{0}$$

